Project Documentation: Design

Yuan Quan - 110212060 - yuan8b@uwindsor.ca

Khadgi, Jiya Kiran- 110229050 - khadgij@uwindsor.ca

Pinzon Niño, Santiago- 110214808 - pinzonn@uwindsor.ca

1. Introduction and Context

1.1 Background and Problem Description

The Canadian retail market has many factors influencing it, including regional economic conditions, industry developments, e-commerce growth, and market competition. However, relevant information is across different public sources and formats, making it tough for business decision-makers to understand market trends. Discrepancies in reporting periods, geographical classifications, industry categories, and data structures also create challenges in comparing information from multiple sources.

The proposed Canadian Retail Market Intelligence System aims to address these challenges by combining retail sales information, economic indicators, and retail competition-related content into a single analytical solution. The system will extract and prepare data from Statistics Canada and the Competition Bureau Canada, organize it in a relational database, and provide analytical results through SQL queries and Power BI visualizations.

The main purpose of the proposed system is to support retail decision-makers in understanding historical retail performance and identifying relevant market trends. Bringing together different sources of data, the proposed solution will provide a more organized approach to examining regional retail performance, industry developments, e-commerce activity, and economic conditions.

1.2 Proposed System Scope

The proposed system will cover the acquisition, preparation, storage, analysis, and visualization of publicly available Canadian retail market data. Its design will consist of the following main activities:

1. Data Acquisition: Collecting structured retail sales data and semi-structured economic indicators from Statistics Canada, as well as unstructured web content from the Competition Bureau Canada.
2. Data Preparation: Using Python to clean, transform, and prepare datasets for analysis, including web scraping and text processing.

3. Database Management: Designing and implementing a relational database using MySQL to store and manage the collected datasets.
4. Data Validation and Integration: Evaluating data quality and integrating relevant information from multiple sources.
5. Data Analysis: Using SQL to examine regional retail performance, industry trends, e-commerce activity, and economic conditions.
6. Data Visualization: Developing Power BI dashboards to present market trends and support business decision-making.

The proposed system will follow a data processing workflow in which data is prepared, stored, analyzed, and presented using Python, SQL, and Power BI.

1.3 Scope Limitations and Out of Scope Activities

The following activities are outside the current project scope:

- Developing a real-time retail sales monitoring platform or live data feed.
- Building predictive machine learning models or automated sales forecasting capabilities.
- Generating automated business recommendations or making decisions on behalf of the users.

The platform results will be based on the available datasets and their reporting periods. The system will support historical analysis and comparisons rather than guarantee predictions about future market conditions.

1.4 Relationship to User Requirements

The proposed design is based on the functional and non-functional requirements established in the User Requirements and Analysis deliverable.

The functional requirements:

FR-01 Store Prepared Datasets

FR-02 Validate Stored Data

FR-03 Retrieve and Filter Retail Sales

FR-04 Analyze Economic Indicators

FR-05 Analyze Retail Competition Information

FR-06 Generate and Present Retail Market Trends

The design will also consider non-functional requirements related to data integrity, data quality, security, performance, reliability, maintainability, and compliance.

NFR-01 Data Integrity

NFR-02 Data Quality

NFR-03 Security

NFR-04 Performance

NFR-05 Reliability

NFR-06 Maintainability

NFR-07 Compliance

These requirements will guide the proposed system structure, database design, data processing activities, and analytical capabilities.

2. Goals and Non-goals

2.1 Goals

The main goal for the proposed Canadian Retail Market Intelligence System is to provide a data-driven solution that is able to organize and analyze data from multiple public sources to support business decision-makers. Focusing on the following objectives:

1. Integrate Multiple Data Sets: Acquire and process structured, semi-structured, and unstructured data from reliable public sources.
2. Maintain Data Quality and Consistency: Clean, transform, validate, and integrate the collected datasets using Python and MySQL.
3. Support Retail Market Analysis: Analyze regional retail performance, industry trends, e-commerce activity, and relevant economic indicators.
4. Provide Accessible Analytical Results: Present analytical findings through SQL queries and Power BI visualizations to support business decisions.

2.2 Non-goals

The following limitations help maintain a clear project scope and avoid introducing features that are outside the original proposal.

The system will not aim to:

- Provide real-time updates.
- Predict future retail sales or market performance.
- Automatically make recommendations.
- Replace strategic business decision-making.

By maintaining these goals and non-goals, the project will remain focused on its primary purpose.

2.3 Measurable Success Criteria

The proposed system will be considered successful when it satisfies the following design and functional criteria.

Objective	Success Criteria
Data Storage	All the prepared datasets are successfully mapped to their corresponding MySQL tables.
Data Validation	The number of records imported into each table matches the number of valid records in its corresponding cleaned CSV files.
Retail Sales Analysis	SQL queries successfully retrieve and filter retail sales records by reporting date, geographic location, and industry.
Economic Analysis	SQL queries retrieve selected economic indicators and allow comparisons across available reports.
Competition Analysis	Users can retrieve article information and compare keyword counts and frequencies across the collected documents.
Data Visualization	Power BI presents selected retail market trends, and the displayed values are consistent with the validated database records and SQL results.
Data Integrity and Security	Primary keys and constraints are applied, and database access and credentials are appropriate.

3. System Architecture

3.1 System Overview

The Canadian Retail Market Intelligence System is designed to collect, process, store, and analyze retail market information from multiple publicly available Canadian data sources. The system follows a layered architecture that separates data acquisition, data preparation, database management, analysis, and visualization.

The main data sources include Statistics Canada and the Competition Bureau Canada. The collected information consists of structured retail sales data, semi-structured economic indicators, and unstructured retail-related articles.

Python is used to collect, clean, transform, and validate the datasets before they are imported into a MySQL relational database. SQL queries are then used to retrieve and analyze the stored information, while Power BI serves as the planned visualization platform for presenting market trends to business users.

3.2 Main System Components

The system consists of five main components:

1. Data Sources: Provide retail sales records, economic indicators, and retail-related articles from publicly available Canadian sources.
2. Data Acquisition and Preparation: Python scripts collect data in different formats, including CSV, JSON, and web content. The scripts clean missing or inconsistent values, standardize data formats, and prepare the datasets for database import.
3. Database Management: MySQL stores the processed information in related tables. Primary and foreign keys are used to connect geographic locations, industries, economic indicators, articles, and keywords while maintaining data consistency.
4. Data Analysis: SQL queries retrieve and summarize the stored data to support regional retail comparisons, industry performance analysis, economic indicator analysis, and retail competition keyword analysis.
5. Data Visualization: Power BI is the planned presentation platform for displaying analytical results through charts and dashboards, allowing business users to explore retail market trends.

3.3 System Interactions

The components interact through a sequential data processing workflow. Data is first collected from external sources and processed using Python. After cleaning and validation, the prepared datasets are imported into MySQL.

The database organizes the information into related tables, allowing SQL queries to retrieve and combine relevant records. The analytical results can then be used in Power BI to create visualizations for retail strategy managers, business analysts, and other decision-makers.

This architecture separates data processing, storage, analysis, and presentation into distinct components. As a result, individual components can be updated or maintained with less impact on the overall system.

The detailed logical architecture, physical architecture, and database relationships are presented in Sections 4, 5, and 6, respectively.

4. Logical Architecture

4.1 Overview

The logical architecture describes how the proposed Canadian Retail Market Intelligence System is organized into functional layers, independent of the hardware or tools used to run them. It follows the data processing workflow described in Section 1.2, in which data is prepared, stored, analyzed, and presented using Python, SQL, and Power BI. Each layer has one main responsibility and passes its output to the next layer.

4.2 Logical Layers

The proposed system consists of the following layers:

- **Data Source Layer:** Contains the publicly available data used by the system: structured retail sales data from Statistics Canada, semi-structured economic indicators from Statistics Canada, and unstructured web content from the Competition Bureau Canada.
- **Data Acquisition and Preparation Layer:** Uses Python to collect the source data, perform web scraping and text processing, and clean, transform, and validate the datasets. The output of this layer is a set of cleaned CSV files.
- **Database Layer:** Stores the integrated datasets in a MySQL relational database. Primary keys and constraints are applied to maintain data consistency and prevent duplicate records.
- **Analysis Layer:** Uses SQL queries to retrieve and examine retail sales, economic indicators, and retail competition information.
- **Presentation Layer:** Uses Power BI dashboards to present retail market trends to decision-makers.

4.3 Relationship Between Layers and Requirements

Layer	Main Function	Related Requirements
Data Source	Provides public retail, economic, and competition data	NFR-07 Compliance
Data Acquisition and Preparation	Collects, cleans, and validates data	NFR-02 Data Quality
Database	Stores prepared datasets and applies keys and constraints	FR-01 Store Prepared Datasets, FR-02 Validate Stored Data, NFR-01 Data Integrity, NFR-03 Security

Analysis	Retrieves and compares data using SQL	FR-03 Retrieve and Filter Retail Sales, FR-04 Analyze Economic Indicators, FR-05 Analyze Retail Competition Information, NFR-04 Performance
Presentation	Displays trends in Power BI	FR-06 Generate and Present Retail Market Trends

4.4 Data Flow

Data moves through the layers in one direction. Data from Statistics Canada and the Competition Bureau Canada is collected and cleaned with Python, stored as cleaned CSV files, and loaded into the corresponding MySQL tables. The number of records imported into each table is then checked against the number of valid records in its cleaned CSV file. SQL queries analyze the stored data, and Power BI presents the results. Consistent with the scope limitations in Section 1.3, the system supports historical analysis and does not include real-time data feeds, predictive models, or automated recommendations.

Because each layer has a single responsibility, a change to one layer, such as adding a dataset, does not require changes to the others. This supports NFR-06 Maintainability and NFR-05 Reliability.

5. Physical Architecture

5.1 Overview

The physical architecture describes the tools and platforms on which the logical layers of the proposed system are deployed. The system uses Python for data processing, MySQL for data storage, Power BI for visualization, and GitHub for code storage and collaboration.

5.2 Physical Components

Component	**Role**	**Technology**
Data Processing Environment	Runs scripts for data collection, web scraping, cleaning, and validation	Python
Database	Stores the integrated datasets and supports SQL analysis	MySQL

Visualization Platform	Presents retail market trends	Power BI
Code Repository	Stores code, datasets, database schema, and documentation	GitHub
External Data Sources	Provide the public data	Statistics Canada and Competition Bureau Canada websites

5.3 Component Descriptions

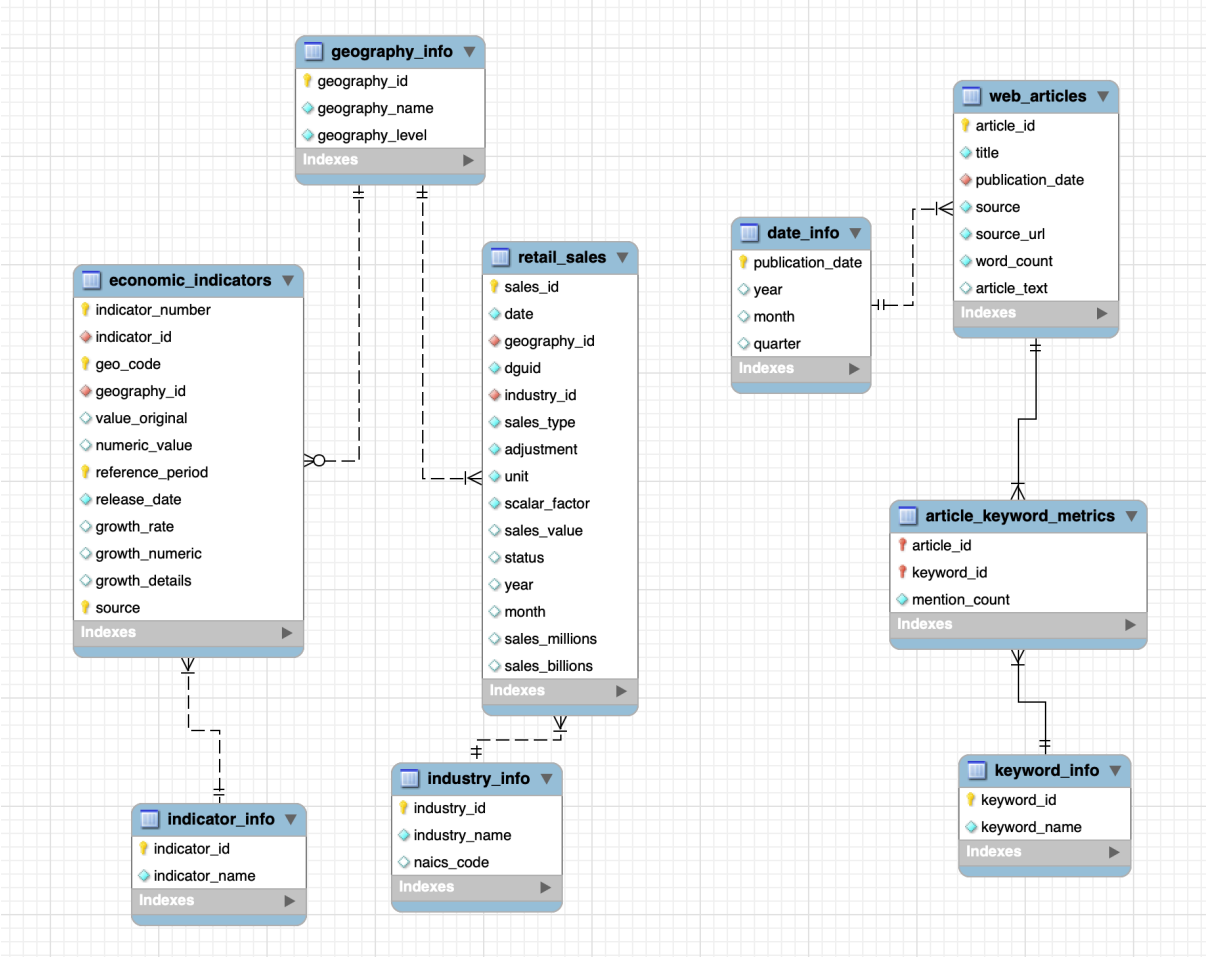
- **Data Processing Environment:** Python scripts collect data from the external sources and prepare it for storage. This is the only part of the system that connects to the external sources.
- **Database:** The prepared datasets are loaded into a MySQL database. SQL queries are run in MySQL to analyze the data. Database access is limited to authorized team members to support NFR-03 Security.
- **Visualization Platform:** Power BI is used to present the analytical results as dashboards, so users can view market trends without writing SQL queries.
- **Code Repository:** GitHub is used for code storage, version control, and team collaboration. The repository holds the source code, dataset files, database schema, and project documentation, which supports NFR-06 Maintainability.
- **External Data Sources:** The system connects to Statistics Canada and the Competition Bureau Canada only to collect publicly available data. After the data is stored in MySQL, analysis and visualization do not depend on live access to these sources. This supports NFR-07 Compliance.

5.4 Deployment Flow

Python scripts collect and prepare the data, the cleaned data is loaded into the MySQL database, SQL queries analyze the stored data, and Power BI presents the results. Code and documentation are kept in the GitHub repository throughout this process.

6. Data Model

6.1 Entity-relationship diagrams (ERDs)



The ERD shows seven one-to-many relationships between the tables.

The geography_info table is connected to both retail_sales and economic_indicators through geography_id. Each geographic location can have multiple retail sales records and economic indicator records. This allows the two datasets to be analyzed by region.

The industry_info table is connected to retail_sales through industry_id. One industry can have multiple sales records, allowing users to compare retail performance across different industries.

Similarly, indicator_info is connected to economic_indicators through indicator_id. Each indicator type, such as the Consumer Price Index or unemployment rate, can have multiple observations.

For the article data, date_info is connected to web_articles through publication_date. This relationship allows articles to be grouped and filtered by publication date.

The article_keyword_metrics table connects web_articles and keyword_info. Each article can contain multiple keywords, and the same keyword can appear in different articles. The table records how many times each keyword appears in an article.

6.2 Schema Designs

The Canadian Retail Market Intelligence System uses a relational database with nine tables. The database separates retail sales, economic indicators, and web articles from their related reference tables. This structure helps reduce duplicated information, maintain data consistency, and support SQL analysis.

The table below summarizes the database schema, including the primary keys, foreign keys, key attributes, and purpose of each table.

Primary and Foreign Key Design

Each table uses a primary key to identify its records. Most tables use a single-column primary key, while economic_indicators and article_keyword_metrics use composite primary keys.

The economic_indicators table uses four columns as its primary key: source, indicator_number, geo_code, and reference_period. Together, these columns identify each economic indicator record by its source, indicator number, geographic code, and reporting period.

Foreign keys connect the main data tables to their reference tables. For example, retail_sales uses geography_id and industry_id to reference geographic locations and retail industries. Similarly, economic_indicators uses geography_id and indicator_id to connect economic observations with their locations and indicator types.

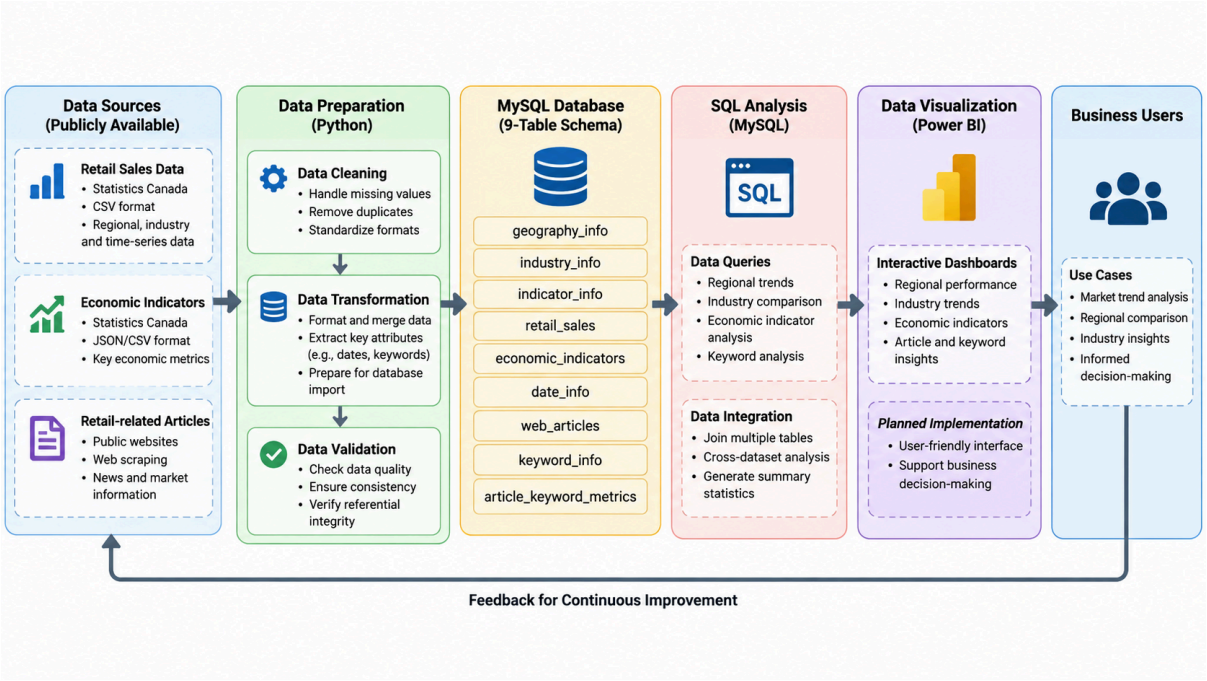
The article_keyword_metrics table connects web_articles and keyword_info through article_id and keyword_id. These two columns also form its composite primary key, preventing duplicate combinations of the same article and keyword.

Overall, the schema is designed to keep related data organized, reduce redundancy, and make it easier to retrieve and compare information across different datasets.

Table Name	Primary Key	Foreign Keys	Key Attributes	Purpose
geography_info	**geography_id**	**—**	**geography_name, geography_level**	**Stores geographic locations and their classifications.**
industry_info	**industry_id**	**—**	**industry_name, naics_code**	**Stores retail industry categories and their NAICS codes.**
indicator_info	**indicator_id**	**—**	**indicator_name**	**Stores different economic indicator types, such as CPI and unemployment rate.**
retail_sales	**sales_id**	**geography_id, industry_id**	**date, sales_value, sales_type, adjustment, year, month**	**Stores retail sales records by geographic location, industry, and reporting period.**

economic_indicators	**(source, indicator_number, geo_code, reference_period)**	**geography_id, indicator_id**	**numeric_value, release_date, growth_rate, growth_numeric**	**Stores economic indicator observations across geographic locations and reporting periods.**
date_info	**publication_date**	**—**	**year, month, quarter**	**Stores publication dates and related calendar information.**
web_articles	**article_id**	**publication_date**	**title, source, source_url, word_count, article_text**	**Stores retail-related articles and their publication information.**
keyword_info	**keyword_id**	**—**	**keyword_name**	**Stores keywords used to analyze retail competition and market trends.**
article_keyword_metrics	**(article_id, keyword_id)**	**article_id, keyword_id**	**mention_count**	**Records how often each keyword appears in an article.**

6.3 Data flow charts



The data flow diagram shows how data moves through the system, from collection to reporting.

Retail sales data, economic indicators, and retail-related articles are collected from publicly available Canadian sources. Python is used to clean, transform, and validate the data before it is imported into MySQL.

The database stores the information in related tables, allowing SQL queries to retrieve and compare data across regions, industries, and economic indicators. The results will then be used in Power BI to create visualizations that support business analysis and decision-making.

The system is divided into four main components: data collection, data preparation, database management, and data analysis and visualization. Each component has a specific role in processing retail market information.

7. Component Design:

7.1 Data Collection

The data collection component gathers information from three sources. Retail sales data and economic indicators are obtained from Statistics Canada in CSV or JSON format.
Retail-related articles are collected from publicly available websites.

The collected data is saved in its original format before further processing. This allows the team to check the original records if any issues are found during data cleaning or analysis.

7.2 Data Preparation

Python is used to clean and prepare the collected datasets before importing them into MySQL. The main tasks include checking missing values, removing duplicate records where appropriate, standardizing data formats, and converting numerical values into suitable data types.

For article data, the program also counts selected keywords related to retail competition, pricing, grocery, consumers, and online shopping. The number of mentions per 1,000 words is calculated to make articles of different lengths easier to compare.

The general processing logic is:

1. Load the raw dataset.
2. Check for missing or duplicate values.
3. Standardize column names and data formats.
4. Calculate additional fields where needed.
5. Validate the processed records.
6. Export the cleaned data for database import.

Article Keyword Frequency Calculation

INPUT: Article text, selected keywords

FOR each article:

    Count the total number of words

    FOR each selected keyword:

        Count keyword occurrences

        IF total word count > 0:

            Frequency = (Occurrences / Total Words) * 1000

        ELSE:

            Frequency = 0

        Store article ID, keyword ID,

        occurrence count and frequency

OUTPUT: Article keyword statistics

7.3 Database Management

MySQL stores the processed data in nine related tables. The database uses primary keys to identify records and foreign keys to connect related information.

For example, when a retail sales record is imported, its geographic location and industry are matched with the corresponding records in geography_info and industry_info. The related IDs are then stored in retail_sales.

A similar process is used for economic indicators. The article keyword data is stored separately in article_keyword_metrics, which connects each article with its keywords and mention counts.

7.4 Data Analysis and Visualization

SQL queries are used to retrieve and summarize information from the database. The main analysis tasks include comparing retail sales across regions and industries, examining economic indicators, and identifying frequently mentioned topics in retail-related articles.

The planned Power BI component will present selected query results through charts and tables. Users will be able to compare market trends and review information relevant to business decisions.

8 Interface Design

8.1 Wireframes and Screen Layouts

The system's user interface will be developed through Microsoft Power BI dashboards. These dashboards will present the results of the data analysis in a visual format, allowing users to explore Canadian retail performance, regional differences, e-commerce activity, economic indicators, and related information.

The proposed draft dashboard is organized into three main pages:

1. Canadian Retail Market.
2. Regional and E-commerce Analysis.
3. Economic and Competition Insights.

The dashboards will use charts, KPI cards, tables, and filters to make information easier to interpret. A consistent layout, including titles, labels, and formatting will be maintained across the pages.

Figure 1. Canadian Retail Market

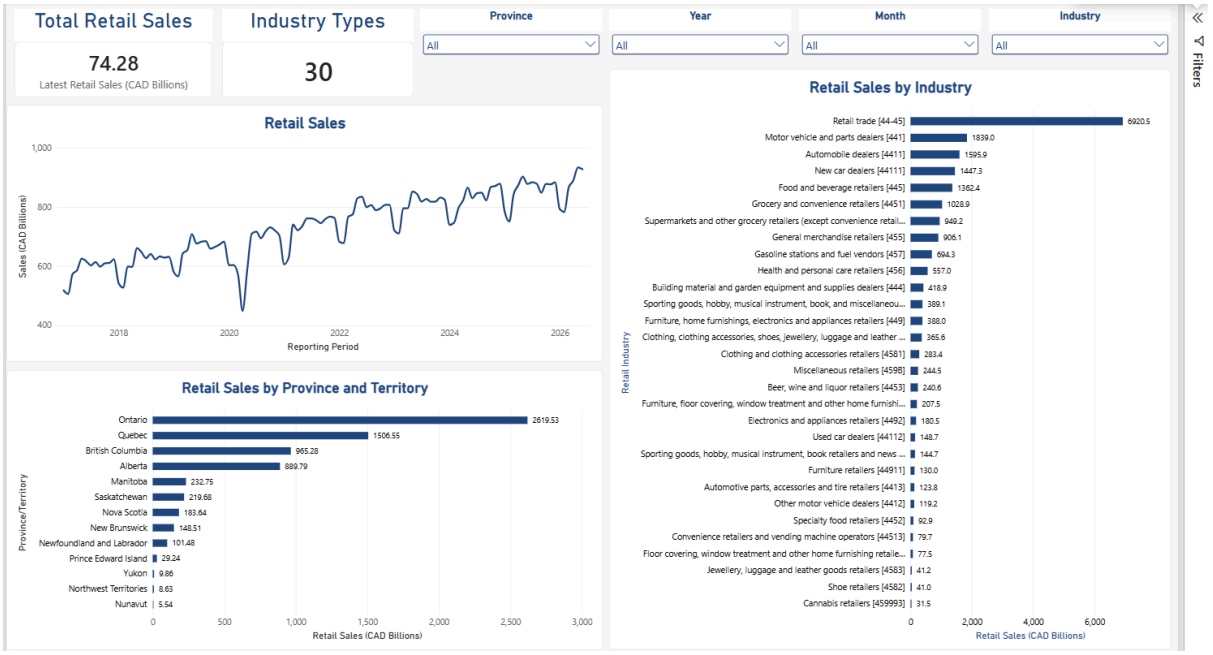


Figure 2. Regional and E-commerce Analysis.

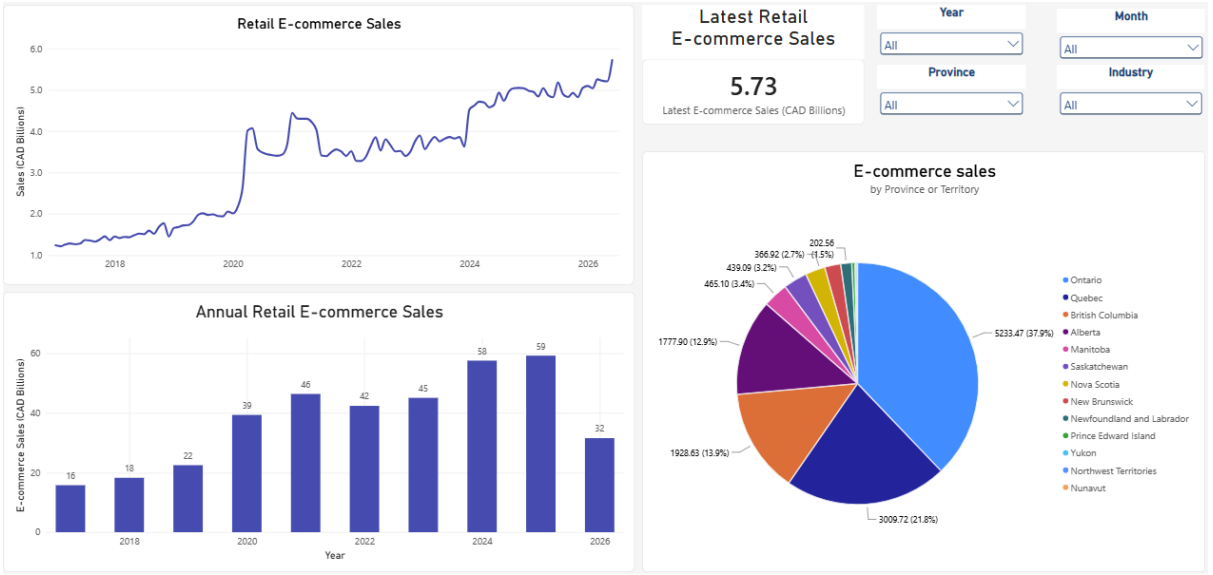
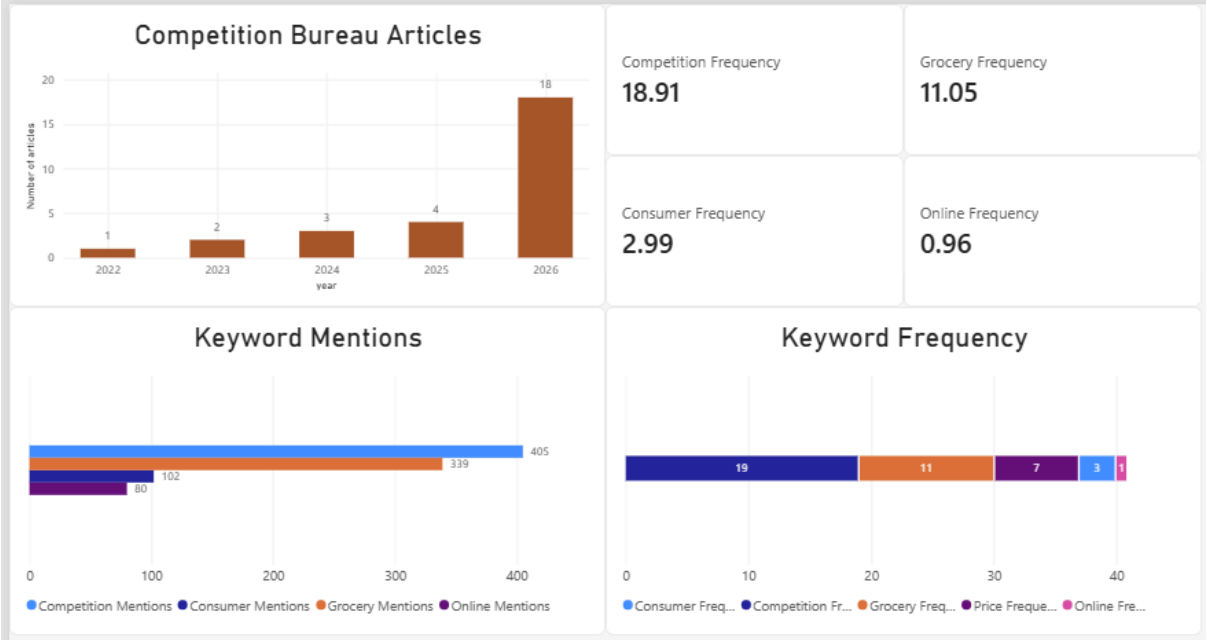


Figure 3. Economic and Competition Insights.



8.2 Protocol Definitions

The data will be transferred and processed using the formats and tools defined in the proposed system architecture.
The main exchange methods are:

- Statistics Canada: Data obtained in CSV and JSON format.
- Competition Bureau Canada: Public webpage content collected using Python.
- Python and MySQL: Python scripts process, clean, transform, and validate the datasets before they are imported into MySQL relational database.
- MySQL and PowerBI: SQL queries retrieve and prepare analytical results for visualization in Power BI.

These methods provide the data exchange required by the project without introducing an additional API layer.

8.3 Constraints and Performance Considerations

The system is subject to several constraints related to data availability, processing, and visualization.

- Historical data availability: The project uses publicly available historical datasets. It does not include real time data collection or automatic updates.
- Different data structures: The three sources contain structured, semi-structured, and unstructured information. Their fields, formats, geographic coverage, and reporting periods must be considered during integration.

- Data Coverage: Economic indicators are available for a limited number of measures and reference periods, while the Competition Bureau dataset contains a limited selection of articles. Therefore, the results should be interpreted within the available data coverage.
- Data comparability: Retail sales and economic indicators cannot always be compared directly because they may use different geographical levels, reporting periods, and measurement units.
- Processing and visualization: The whole process depends on the available computer resources. The team will aim to keep data processing and queries efficient, particularly when filtering and summarizing sales records.

Performance is expected to be evaluated through the successful execution of data imports and analytical queries, as well as the dashboard responsiveness.

8.4 Security Considerations

The system uses publicly available datasets and does not require the collection of personal customer information. However, database access and project files must still be managed securely.

Access to the MySQL database will be restricted to authorized team members. Database credentials should be stored securely and excluded from the public GitHub repository.

Primary and foreign key constraints will be used to maintain data integrity and prevent invalid relationships between tables. Data processing scripts and database operations should also be reviewed to reduce the risk of accidental data modification.

These measures support the system’s security and data integrity requirements.

9. Design Rationale

The project uses Python, MySQL, and Power BI because each tool supports a different stage of the data analysis process.

9.1 Python for Data Processing

Python was selected for data collection and preparation because it can work with different data formats, including CSV, JSON, and web content. It also provides libraries for cleaning data, handling missing values, and performing calculations.

Using Python allows the team to prepare the datasets before importing them into MySQL, rather than making all data changes directly in the database.

9.2 MySQL for Database Management

MySQL was chosen because the project involves several datasets that need to be stored and analyzed together. A relational database makes it possible to organize the information into separate tables and connect them using primary and foreign keys.

The database uses supporting tables for geographic locations, industries, and economic indicator types to reduce repeated information. The article keyword data is also separated from the main article table so that additional keywords can be included without changing the structure of web_articles.

This design supports data consistency and makes SQL queries easier to organize.

9.3 Power BI for Data Visualization

Power BI was selected as the planned visualization tool because it can present data through interactive charts, tables, and dashboards.

For this project, Power BI will be used to display regional retail performance, industry comparisons, economic indicators, and article keyword trends. These visualizations are intended to make the results easier for business users to understand.

9.4 Overall Design Approach

The system follows a step-by-step process, starting with data collection and preparation, followed by database storage, SQL analysis, and visualization.

Keeping these tasks separate makes the project easier to manage. It also allows the team to update the data preparation process or modify SQL queries without redesigning the entire system.